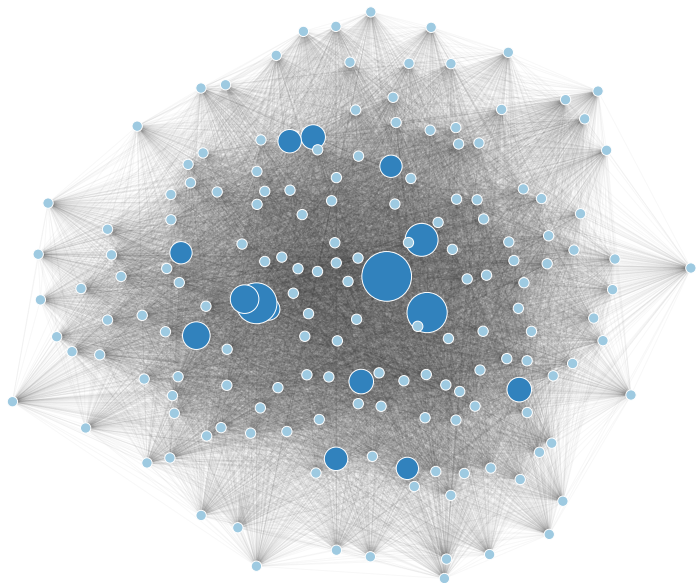


Standard Relationship Inference

$O(N^2)$ All-to-All Complexity

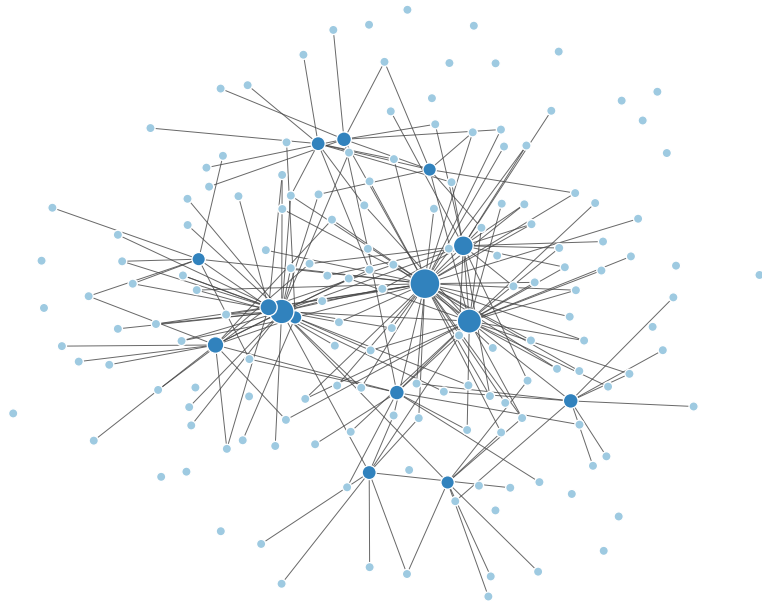


*Redundancy in 170-node system
High Computational Barrier*



RIS Framework

$O(N \log N)$ Adaptive Sparsification



**Topological Preservation
99% Gain in Efficiency**